

An aerial photograph of a city, likely Zurich, showing a river (Limmat) flowing through the center. The river is surrounded by dense urban development, including various buildings and bridges. The image is used as a background for the slide.

# Project: Privacy-Preserving Federated Learning

From Data to Solutions  
ETH Zürich

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# Recap: What you have learned.



CPU GPU TPU

same job, more units doing it

## CPU, GPU, TPU

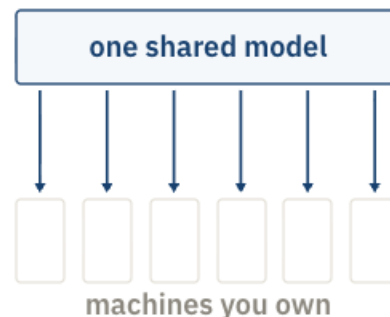
Where the work runs, and why the fastest chip sits idle.



you price the run before you start it

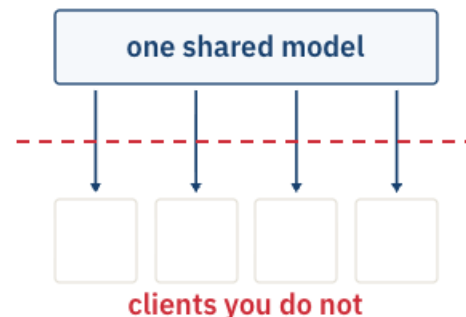
## Training budgets

What a run costs, worked out before it starts.



## Distributed training

One model, many machines. All of them yours.



## This project

Each client has private data you cannot see.

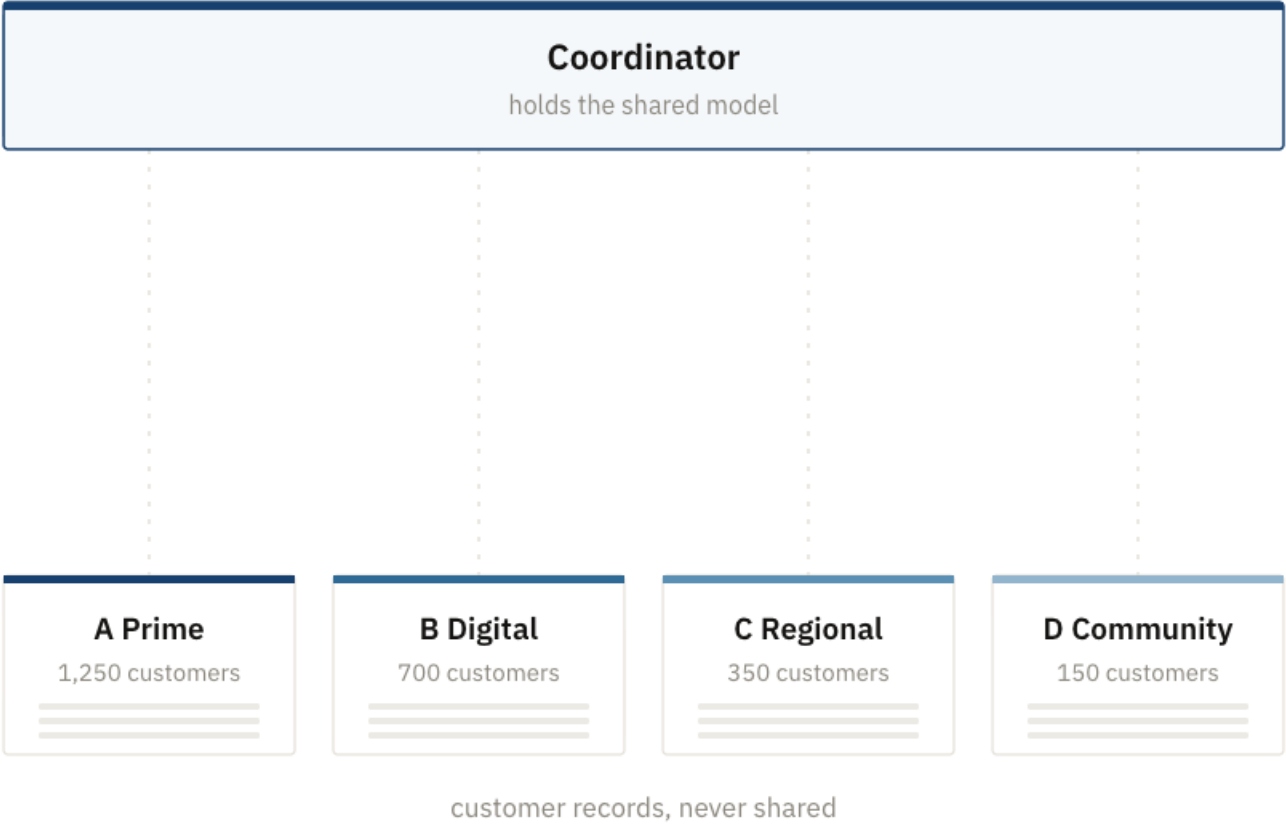
The wall is no longer hardware. It is that the data cannot move at all.



# Federated Learning



# Step 1: Set up the federation.

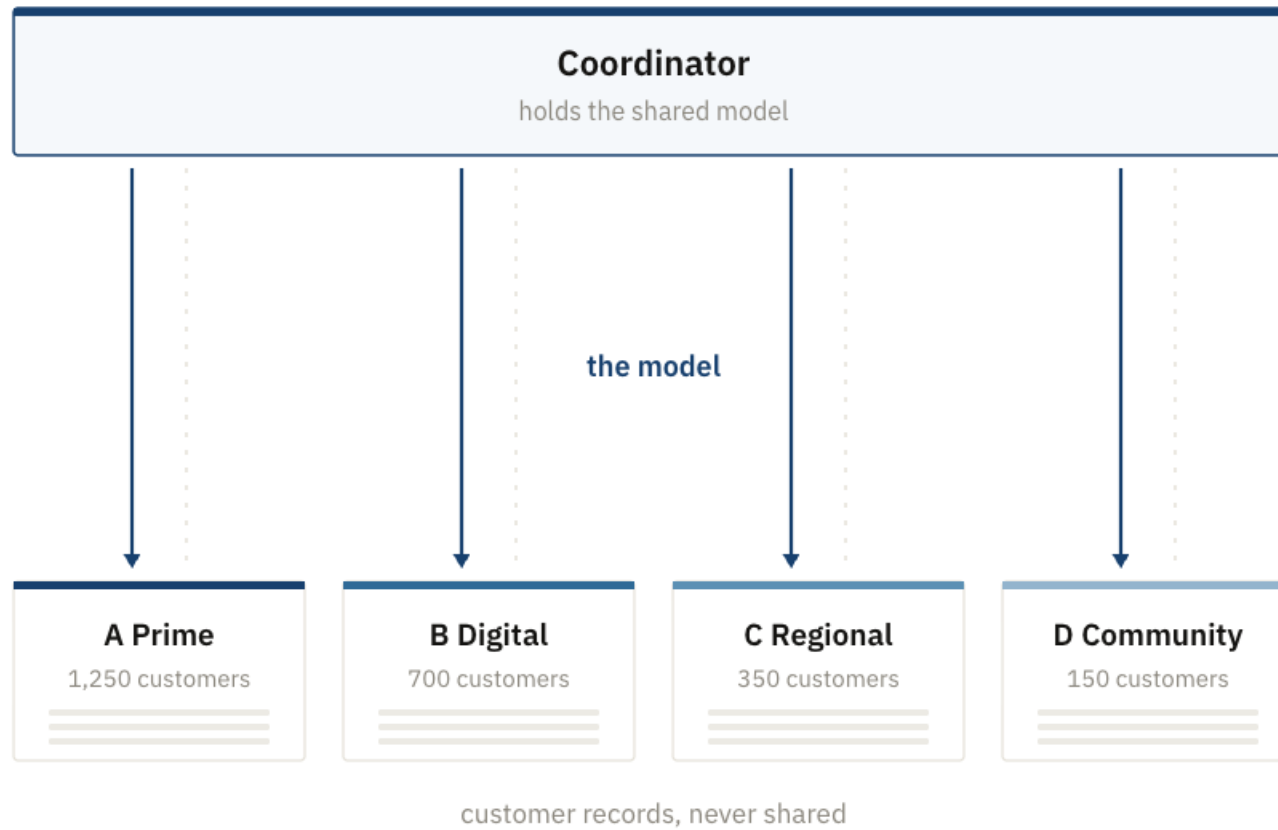


## 1 Set up the federation.

### Real life examples

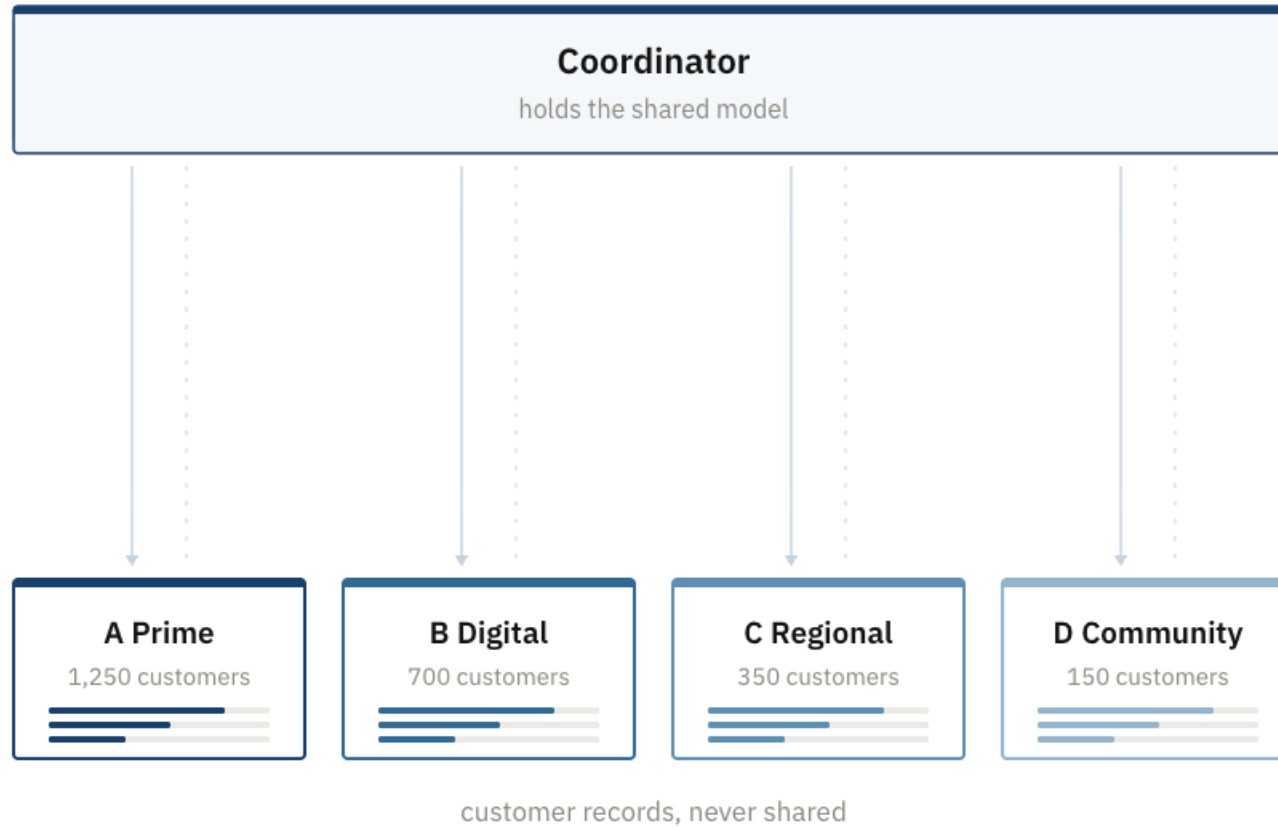
| coordinator       | clients       |
|-------------------|---------------|
| A research centre | Hospitals     |
| A keyboard maker  | Phones        |
| A manufacturer    | Its factories |

## Step 2: The model goes out.



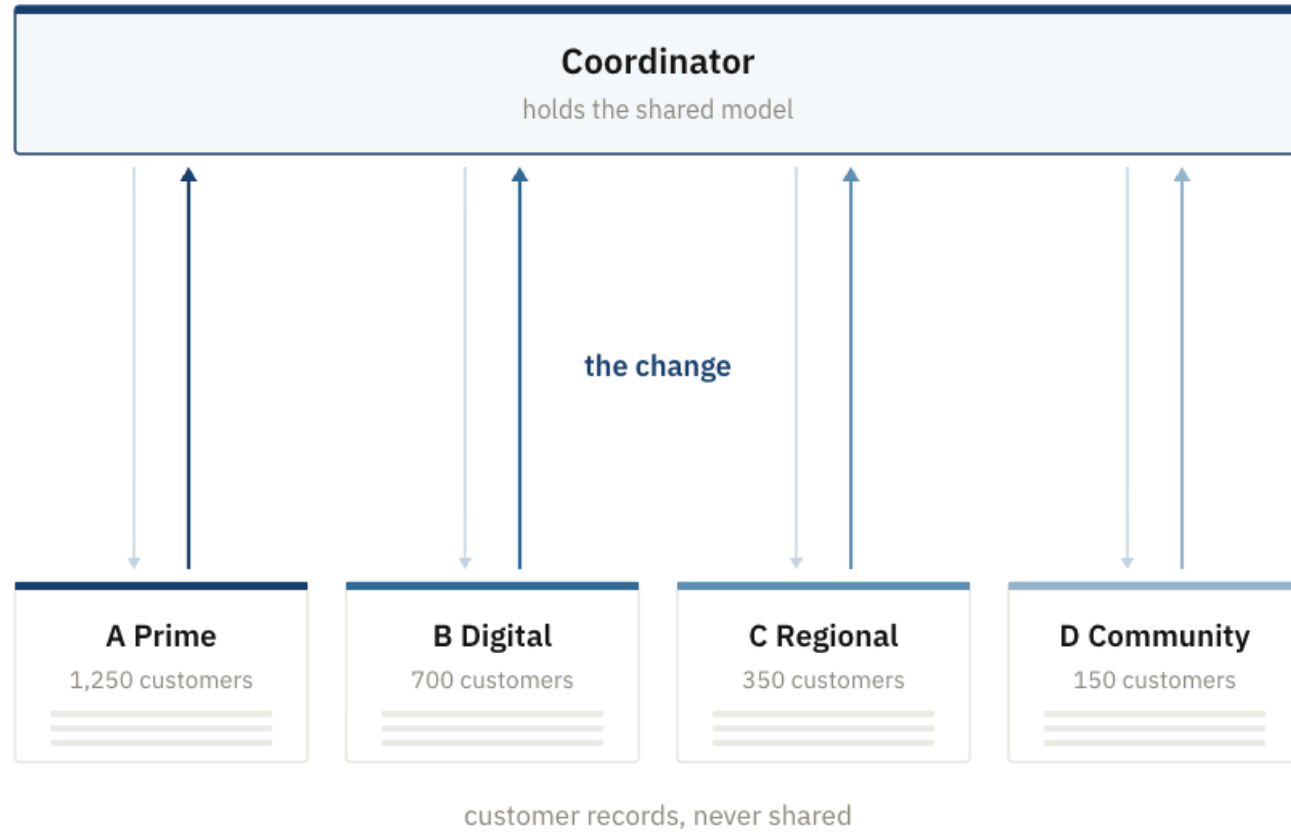
- 1 Set up the federation.
- 2 **The model goes out.**

# Step 3: Each client trains at home.



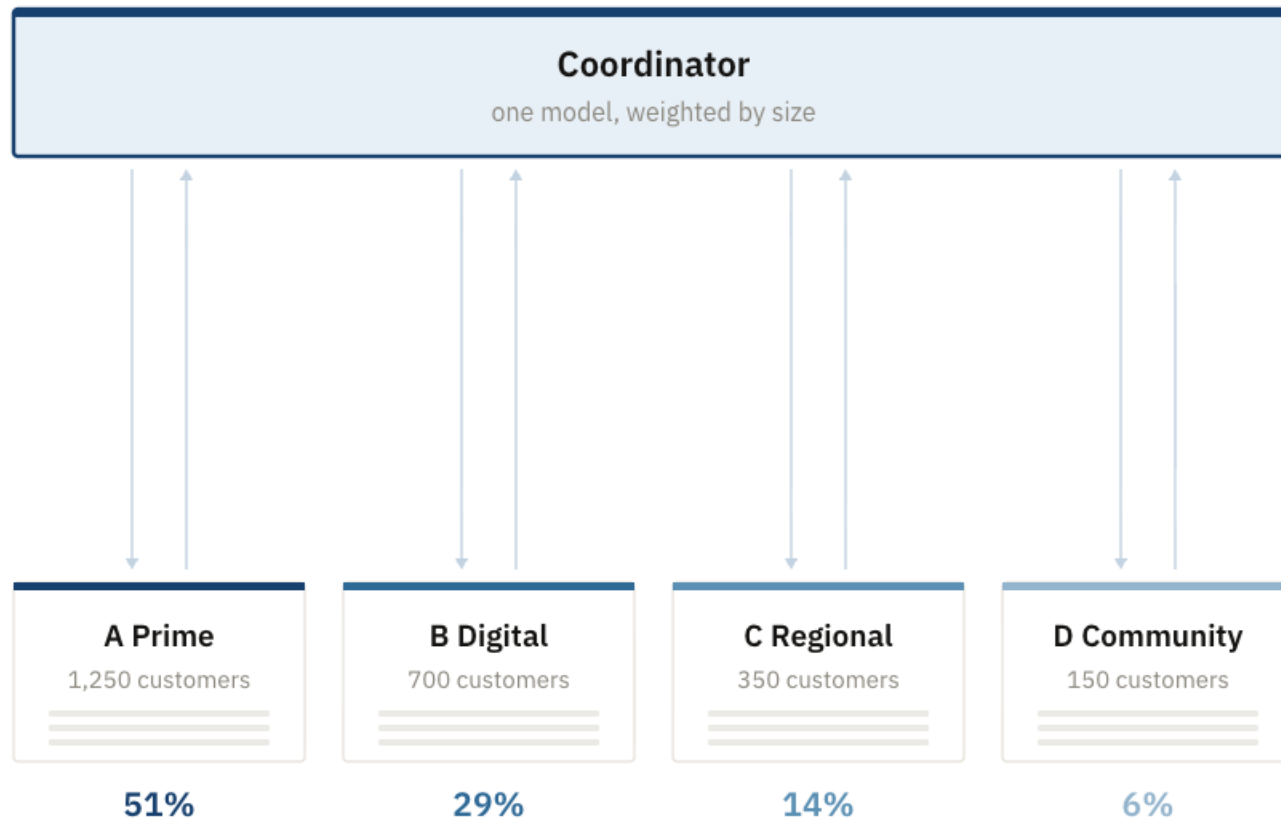
- 1 Set up the federation.
- 2 The model goes out.
- 3 Each client trains at home.**

# Step 4: The change comes back.



- 1 Set up the federation.
- 2 The model goes out.
- 3 Each client trains at home.
- 4 **The change comes back.**

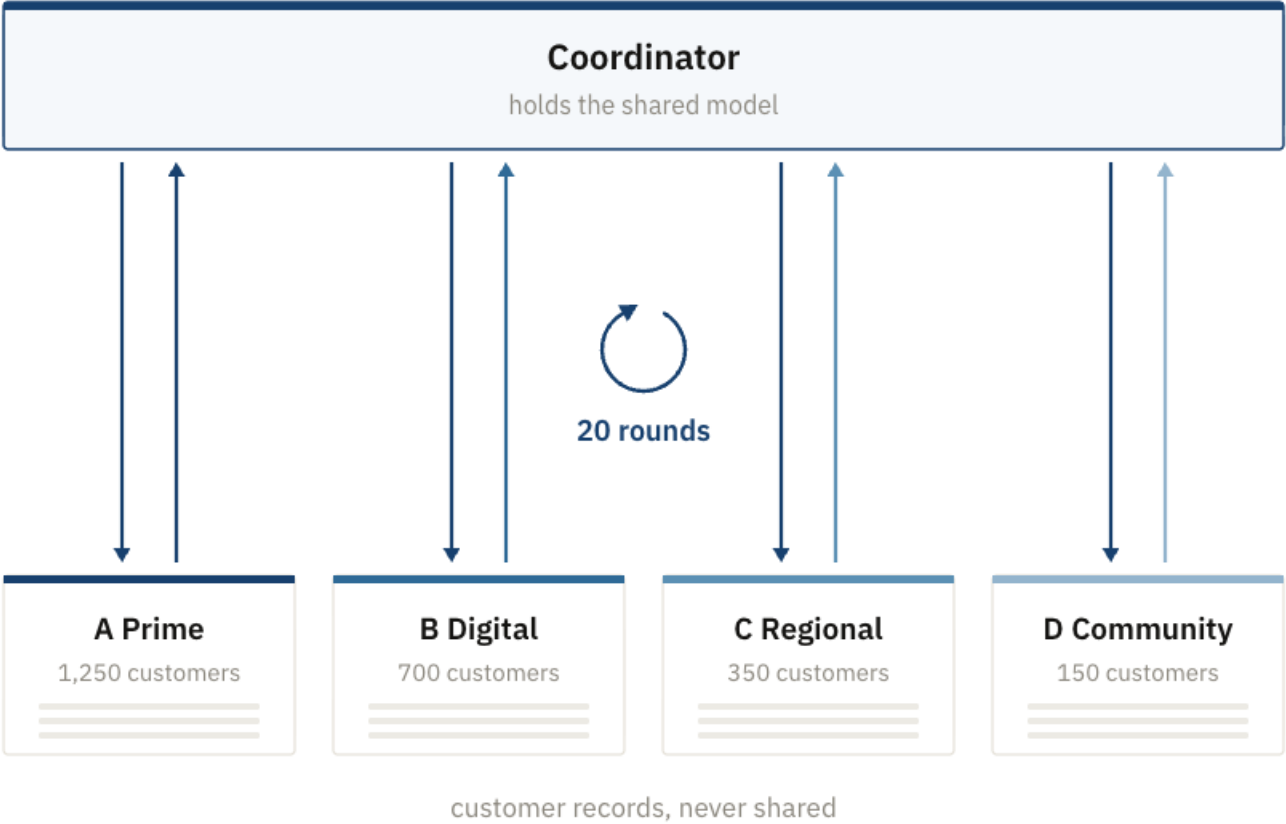
# Step 5: The coordinator averages.



- 1 Set up the federation.
- 2 The model goes out.
- 3 Each client trains at home.
- 4 The change comes back.
- 5 **The coordinator averages.**

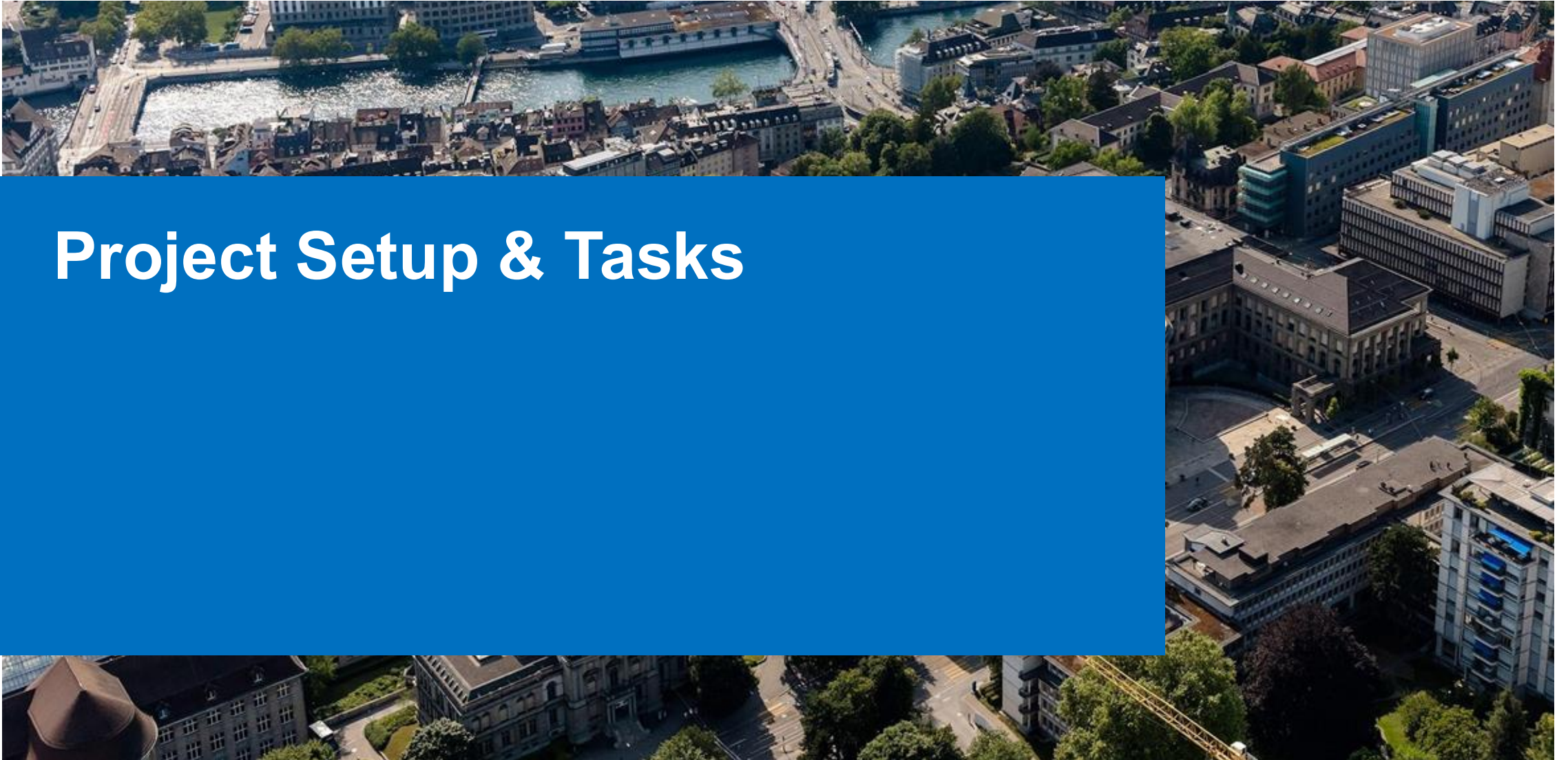


# Steps 2 to 5, again and again.

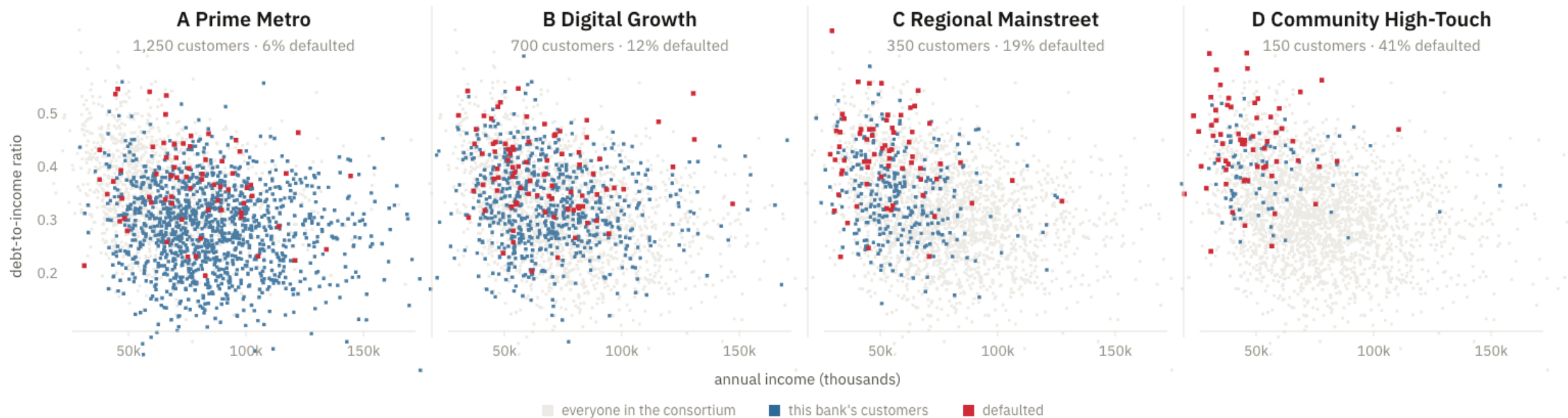


- 1 Set up the federation.
- 2 **The model goes out.**
- 3 **Each client trains at home.**
- 4 **The change comes back.**
- 5 **The coordinator averages.**

# Project Setup & Tasks



# Our Scenario. Our Data.



| Bank                   | Customers                    | Share who defaulted          | Median annual income       |
|------------------------|------------------------------|------------------------------|----------------------------|
| A Prime Metro          | <div><div></div></div> 1,250 | <div><div></div></div> 5.5%  | <div><div></div></div> 82k |
| B Digital Growth       | <div><div></div></div> 700   | <div><div></div></div> 12.3% | <div><div></div></div> 66k |
| C Regional Mainstreet  | <div><div></div></div> 350   | <div><div></div></div> 19.1% | <div><div></div></div> 54k |
| D Community High-Touch | <div><div></div></div> 150   | <div><div></div></div> 40.7% | <div><div></div></div> 45k |
| All four together      | 2,450                        | 11.6%                        | 71k                        |

# Build it, price it, attack it, defend it.

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Parts 1 and 2

Motivation.

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Part 3

**Build it.**

One model across all four. No record ever moves.

Part 4

**Price it.**

What it costs in rounds, local work, and bytes.

Part 5

**Attack it.**

Turn one update back into the customer behind it.

Part 6

**Defend it.**

Clip, add noise, and price the promise that buys.

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Part 7, optional

What changes when the model, the federation, or the payload gets bigger.

# The hand-in is the notebook itself.

4 · Now run the attack yourself

Task 4 did the reconstruction in Python, on an update the cell handed you. This does it on the wire, on an update you cause a bank to send.

**Heist.** Set one bank to train on a single customer, run a round, then open that bank's  $\Delta w$  on the coordinator's table and divide. The mission is achieved when what comes back matches a real customer in that bank's file.

Same frame as before: scroll inside it, and press **Back to the notebook** when the card appears.

```
1 #title 5.4 - mission: the heist (run me, then play) { display-mode: "form" }
2 fv.mission("heist")
```

Heist — Turn a dial until one bank's update hands back a real customer.

One model, four banks

Heist

Mission 4 - In progress

An attack on the wire that comes back with an exact match.

125 pts

Review concept

A Prime Metro

1,250 customers, 51% of the average

Joins

Local steps

Batch / step

in

5

all

B Digital Growth

700 customers, 29% of the average

Joins

Local steps

Batch / step

in

5

all

C Regional Mainstreet

350 customers, 14% of the average

Joins

Local steps

Batch / step

in

5

all

D Community High-Touch

150 customers, 6% of the average

Joins

Local steps

Batch / step

in

5

all

Coordinator

untrained

$\times 51\%$

$\times 29\%$

$\times 6\%$

$\times 14\%$

Nothing is travelling yet. The shared model is untrained.

Animation

slow

normal

fast

Shared model score, untrained

0.500

pooled 0.850

Pooled reference

0.850

NEW CONTROL - ATTACK SURFACE

How many customers shape an update?

Batch / step 1 32 all

Customers sampled in one local step. A larger batch mixes more people, but during private training it also touches more of the book and raises  $\epsilon$ .

Clipping, noise, and privacy scores are deliberately unavailable here. First expose why a one-customer update is dangerous.

MISSION 4

Heist

You are the coordinator, and you are curious. Turn a dial until one bank's update hands back a real customer. Run a round, then select one returned gradient on the coordinator's table to attack it.

What it takes

An attack on the wire that comes back with an exact match.

Hint

Hover

## 1 Fill in the code

Six tasks, twelve blanks. Run the cell and it marks each one.

## 2 Play the missions

Three of them, played in the game, which opens inside the notebook.

## 3 Fill in the tables

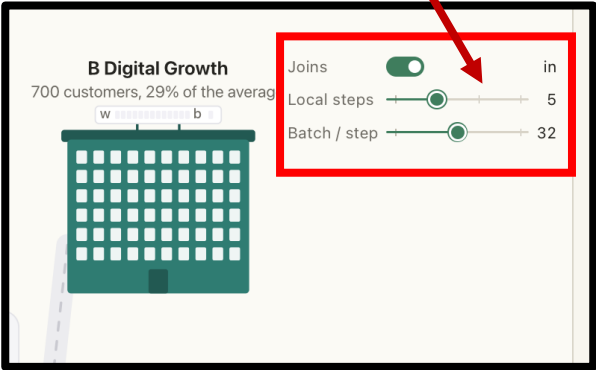
Five short tables, a few lines each.

Nothing else to submit.

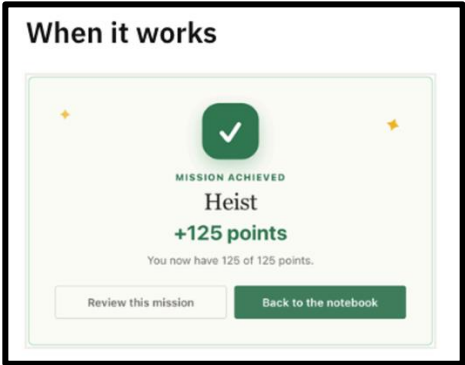


# Mission Interface

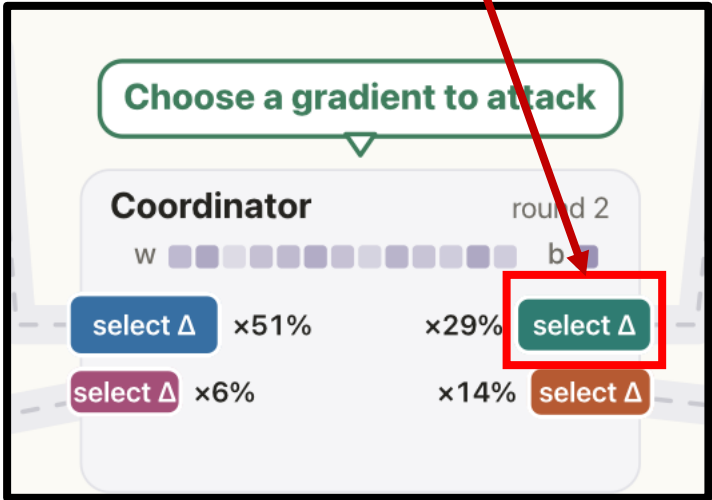
Play with dials.



If you complete a mission, you will get this.



You can select one of these.



There are hints for missions.

